

Non-convex optimal investment with transient price impact

Lóránt Nagy

HUN-REN Rényi Institute and ELTE University, Budapest

Joint work with Miklós Rásonyi

Vienna, VCMF, Tuesday 2nd September, 2025

Illiquidity, and Price impact

- ▶ absence of liquidity
- ▶ biting into the order book leads to less favorable prices
- ▶ buying and selling moves prices against the trader
- ▶ What is the temporal nature of price impact?
 - . permanent, and informational
 - . temporal, and instantaneous
 - . temporal, and transient

Illiquidity, and Price impact

PERMANENT IMPACT – LARGE TRADERS

- ▶ Bank and Baum; 2004

$$S_t = P(\vartheta_t, t); \quad \vartheta_t \sim \text{inventory at time } t$$

- ▶ Bank and Kramkov; 2015
With CARA investor facing Bachelier:

principle of utility indifference prices $\rightarrow$

$$S_t = S_t(\vartheta_t) = \mu_t + \sigma B_t + \lambda \vartheta_t$$

ANALYTICAL TRACTABILITY IS NOT STRAIGHTFORWARD

- ▶ concave, non-decreasing utility – $\rightarrow$ has optimal strategy
- ▶ endogenous price paths, $S = S(\vartheta)$, no "usual" convex analysis toolkit

Illiquidity, and Price impact

INSTANTANEOUS PRICE IMPACT - SMALL TRADERS

- ▶ Guasoni and Rásonyi; 2015

Cádlág price S_t . portfolio position at time t is $\vartheta_t = \int_0^t \phi_u du$.



$$V_t = V_0 - \int_0^t \phi_u S_u du - \int_0^t G_u(\phi) du$$

HAS CONVEX STRUCTURE

- ▶ convex $G_t(\cdot)$; execution price: $S_t + G_t(\phi_t)$ (Komlós's theorem is used)
- ▶ concave, non-decreasing utility $\rightarrow$ has optimal strategy

Illiquidity and Price Impact

TRANSIENT PRICE IMPACT – MID-SIZE TRADERS

- ▶ Bank and Dolinsky; 2019 Bank and Voß; 2018
- ▶ Continuous price S_t ; predictable, càdlàg, BV inventory with $\vartheta = \vartheta_t^\uparrow - \vartheta_t^\downarrow$; *observed* price $S_t^\vartheta = S_t + \iota \vartheta_t$
- ▶ LIQUIDITY PRIMITIVES

$$\delta_t > 0, \quad r_t \geq 0$$

- ▶ SPREAD DYNAMICS

$$\zeta_t^\vartheta = \frac{1}{\delta_t} (d\vartheta^\uparrow + d\vartheta^\downarrow) - r_t \zeta_t^\vartheta dt$$

- ▶ CASH DYNAMICS

$$\xi_T^\vartheta = \xi_0 - \int_0^T S_t^\vartheta \circ d\vartheta_t - \int_0^T \zeta_t^\vartheta \circ d(\vartheta^\uparrow + \vartheta^\downarrow).$$

Illiquidity and Price Impact

TRANSIENT PRICE IMPACT – MID-SIZE TRADERS

$$\delta_t > 0, \quad r_t \geq 0, \quad \rho_t := \exp\left(\int_0^t r_s ds\right)$$

monotonicity assumption

$t \rightarrow \delta_t / \rho_t^2$ is strictly decreasing

- ▶ LIQUIDITY-WEIGHTED SPREAD

$$\eta_t^\vartheta = \rho_t \zeta_t^\vartheta$$

- ▶ COST FUNCTIONAL

$$-\xi_T^\vartheta = \Lambda_T^\vartheta = \int_{[0,T]} S_t d\vartheta_t + \frac{1}{2} \int_{[0,T]} (\eta_t^\vartheta)^2 \mu(dt)$$

- ▶ concave, increasing, C^1 utility – > verification result (heavy convex analysis)

A Discrete Time Transient Price Impact Model

OPTIMALITY IN A NON-CONVEX SETUP – A NOVEL RESULT

MARKET AND TRADING DATES

- ▶ Discrete times $t = 0, 1, \dots, T$ with a fixed horizon $T \geq 2$.
- ▶ $(\Omega, \mathcal{F}, \mathbb{P})$ with discrete filtration $(\mathcal{F}_t)_{t=0}^T$.

PRIMITIVES

- ▶ adapted price process S_t .
- ▶ market depth $\delta_t > 0$ and resilience $r_t \geq 0$ (adapted)
- ▶ resilience factors $\rho_{j,t} := \exp\left(-\sum_{i=j}^{t-1} r_i\right)$, $0 \leq j \leq t \leq T$.

TRADING STRATEGY

- ▶ predictable positions ϑ_t with $\vartheta_0 = 0$ and *mandatory liquidation* $\vartheta_T = 0$.
- ▶ increments $H_t := \Delta\vartheta_t = \vartheta_t - \vartheta_{t-1}$, $1 \leq t \leq T$.

A Discrete Time Transient Price Impact Model

OPTIMALITY IN A NON-CONVEX SETUP – A NOVEL RESULT

SPREAD DYNAMICS

$$\zeta_{t+1}^{\vartheta} = e^{-r_t} \zeta_t^{\vartheta} + \frac{|H_{t+1}|}{\delta_{t+1}}, \quad \zeta_0 \geq 0.$$

CASH ACCOUNT

$$\xi_{t+1}^{\vartheta} - \xi_t^{\vartheta} = -S_{t+1}H_{t+1} - \zeta_{t+1}^{\vartheta} |H_{t+1}|, \quad \xi_0^X = 0.$$

EXPLICIT TERMINAL VALUE

$$\xi_T^{\vartheta} = - \sum_{t=1}^T S_t H_t - \sum_{t=1}^T |H_t| \left(\rho_{0,t} \zeta_0 + \sum_{j=1}^t \frac{\rho_{j,t}}{\delta_j} |H_j| \right).$$

WEALTH AND OBJECTIVE Find ϑ^* so that

$$V_T^{\vartheta} = z + \xi_T^{\vartheta} - B, \quad \sup_{\vartheta \in \mathcal{A}_0} \mathbb{E}[u(V_T^{\vartheta})] = \mathbb{E}[u(V_T^{\vartheta^*})], \quad \text{remark!}$$

A dynamical programming approach

THE BACKWARD INDUCTION.

One-step cash flow at t

For $h_1, \dots, h_{t-1}, h_t$ let

$$\kappa_t(h_1, \dots, h_t) := -S_t h_t - |h_t| \left(\rho_{0,t} \zeta_0 + \sum_{j=1}^t \frac{\rho_{j,t}}{\delta_j} |h_j| \right).$$

One-step problem Fix $x, h_1, \dots, h_{T-1}$

$$J_{T-1}(x, h_1, \dots, h_{T-1}) = \operatorname{ess\,sup}_{H \in \Xi_{T-1}} \mathbb{E} \left[u(x + \kappa_T(h_1, \dots, h_{T-1}, H)) \mid \mathcal{F}_{T-1} \right]$$

- We wish for the existence of $H^* = H^*(x, h_1, \dots, h_{T-1})$ such that

$$\begin{aligned} J_{T-1}(x, h_1, \dots, h_{T-1}) \\ = \mathbb{E} \left[u(x + \kappa_T(h_1, \dots, h_{T-1}, H^*)) \mid \mathcal{F}_{T-1} \right] \end{aligned}$$

A dynamical programming approach

THE BACKWARD INDUCTION.

- Given $H^* = H^*(x, h_1, \dots, h_{T-1})$ so that

$$\begin{aligned} J_{T-1}(x, h_1, \dots, h_{T-1}) &= J_{T-1}(x, h_1, \dots, h_{T-2}, h_{T-1}) \\ &= \mathbb{E} \left[u(x + \kappa_T(h_1, \dots, h_{T-1}, H^*)) \mid \mathcal{F}_{T-1} \right] \end{aligned}$$

Next indirect utility

$$\begin{aligned} U_{T-1}(x, h_1, \dots, h_{T-2}, H) \\ = \mathbb{E} \left[J_{T-1}(x + \kappa_{T-1}(h_1, \dots, h_{T-2}, H), h_1, \dots, h_{T-2}, H) \mid \mathcal{F}_{T-2} \right] \end{aligned}$$

Next value function

$$\begin{aligned} J_{T-2}(x, h_1, \dots, h_{T-2}) \\ = \operatorname{ess\,sup}_{H \in \Xi_{T-2}} U_{T-1}(x, h_1, \dots, h_{T-2}, H) \end{aligned}$$

Unusual features

NON-CONVEX PAYOFF

EXAMPLE 1

- ▶ $T = 3$; $r_t = 0$; $S_0 = \zeta_0 = 0$; $S_1, S_2, S_3 \sim \perp, \mathcal{N}(0, 1)$; $\delta_1 = 1$, $\delta_2 = \delta_3 = 10$; $\vartheta_3 = 0$
- ▶ Consider the set of superhedgable claims

$$\mathcal{S} = \left\{ Y \in L^0 : \exists \vartheta \in \mathcal{A}_0 \text{ with } \xi_T^\vartheta \geq Y \right\}.$$

OBSERVATION

- ▶ $\mathcal{S}$ is *not convex*.

Unusual features

NON-CONCAVE UTILITY

EXAMPLE 2

- ▶ One-step, frictionless market. Admissible positions $\Phi = \{0, 1\}$.
- ▶ Utility $u(x) = \ln x$ for $x > 0$; initial wealth $z > 1$.
- ▶ Return on the risky asset $R = \begin{cases} 2 & \text{with prob. } \frac{1}{2}, \\ -1 & \text{with prob. } \frac{1}{2}. \end{cases}$

INDIRECT UTILITY

$$\begin{aligned}\bar{u}(z) &= \sup_{\phi \in \Phi} \mathbb{E}[u(z + \phi R)] \\ &= \max \left\{ \ln z, \frac{1}{2} [\ln(z + 2) + \ln(z - 1)] \right\}\end{aligned}$$

OBSERVATION

- ▶ At $z = 2$ the two candidates cross, but the right derivative exceeds the left, so $\bar{u}$ fails concavity.

Necessary conditions

History-free upper bound

$$\kappa_t(h_1, \dots, h_t) \leq \lambda_t(h_t) := -S_t h_t - \frac{h_t^2}{\delta_t}$$

Cash-flow bound

$$\kappa_t(h_1, \dots, h_t) \leq \frac{S_t^2 \delta_t}{4}$$

ASSUMPTIONS

Integrability For all $x, y \in \mathbb{R}$ and for all $t = 1, \dots, T$ we have

$$\mathbb{E}[u(x + yS_t - B)] > -\infty.$$

Properties of the utility The function u is non-decreasing with $u(-\infty) = -\infty$ and bounded from above.

The proof – one-step lemmas

1. Assume that there is an **integrable lower bound** of the indirect utility.
2. We have a **history-free upper bound** of κ_t that has $-\infty$ as a limit, and it is bounded.
3. The indirect utility has **monotonicity** in the cash position, and it is **continuous**.
4. There exists a continuous version of the indirect utilities.
5. The indirect utility tends to $-\infty$ when $|H| \rightarrow \infty$.
6. The continuous version preserves monotonicity in cash on a full-measure set.
7. On a compact parameter domain it is enough to optimize on $\{H \in \Xi : |H| \leq K(\partial C)\}$.
8. The essential supremum of the indirect utility admits a continuous version.
9. The essential supremum can take random parameters.
10. There exists an optimizer.
11. When forming the new indirect utility that feeds the next step we have the properties 1, 2, and 3.

The proof

one-step lemmas

SOME TOOLS

Common problem of selecting a convergent subsequence for some set of strategies.

- ▶ Well known: For example if $\liminf_n |\eta^n| < \infty$ holds, then it is possible to construct a convergent subsequence with carefully chosen stopping times.
- ▶ Snell's problem: in Neveu 1975 it is solved using a tool that makes it possible to select a convergent subsequence from an upward directed set.

Remark: Continuity of the indirect utility implies that there is a continuous version of the essential supremum: with adding certain monotonicity assumptions upper semi-continuity is enough.

THANK YOU FOR YOUR ATTENTION!

The model of Bank and Voß (2018)

$$\delta_t > 0, \quad r_t \geq 0, \quad \rho_t := \exp\left(\int_0^t r_s \, ds\right)$$

$$\zeta_t^\vartheta = \frac{1}{\delta_t} (d\vartheta^\uparrow + d\vartheta^\downarrow) - r_t \zeta_t^\vartheta \, dt$$

$$\xi_T^\vartheta = \xi_0 - \int_0^T S_t^\vartheta \circ d\vartheta_t - \int_0^T \zeta_t^\vartheta \circ d(\vartheta^\uparrow + \vartheta^\downarrow).$$